

USN

NMAM INSTITUTE OF TECHNOLOGY, NITTE
Off-Campus Centre of Nitte (Deemed to be University)

Second Semester M.Tech (Structural Engineering) (CBCS) Degree Examinations

May - June 2024

22CST201 – ADVANCED DESIGN OF STEEL STRUCTURES

Duration: 3 Hours

Max. Marks: 100

Note: Answer **Five** full questions choosing **One** full question from each Unit.

Use of IS 800-2007, IS 811 and Steel Tables is permitted.

Assume any missing data suitably and mention the same.

Unit – I

- | | Marks | BT* | CO* | PO* |
|--|-------|-----|-----|-----|
| 1. a) Discuss the types of collapse mechanisms. | 06 | L*2 | 1 | 1 |
| b) Explain the concept of plastic hinge with an example. | 04 | L2 | 1 | 1 |
| c) Evaluate the shape factor of a symmetrical I section about the major axis (ZZ – axis) with flanges of size 140 mm X 16 mm and web of thickness 8.9 mm with overall depth of 400 mm. | 10 | L3 | 1 | 2 |
| 2. a) Discuss the three basic theorems of plastic analysis. | 06 | L2 | 1 | 1 |
| b) Compute the fully plastic moment in the portal frame shown in the Fig. 2 (b). Assume uniform flexural rigidity throughout. | | | | |

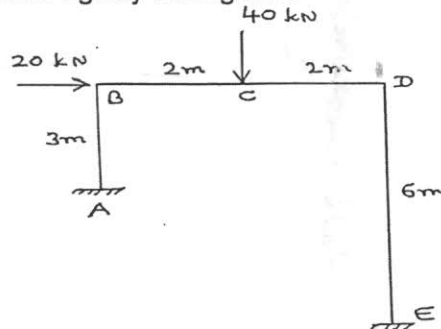


Fig. 2 (b)

14 L3 1 2

Unit – II

- | | | | | |
|---|----|----|---|-----|
| 3. a) Explain the concept of lateral torsional buckling of an I section beam with sketches. | 06 | L2 | 2 | 1,2 |
| b) Evaluate the design bending strength of ISLB 350 @ 486 N/m when used as a laterally unrestrained beam over an unsupported length of 3.0 m. Assume the grade of steel as Fe410. | 14 | L3 | 2 | 1,2 |
| 4. a) State the four types of cross sections used in beam design with the expressions for the design bending strength and stress distribution across the section in each case. | 04 | L2 | 2 | 1 |
| b) Design a laterally unsupported beam simply supported beam of effective span 4.0 m to carry a factored BM of 550 kN-m and factored SF of 200 kN. Assume Fe410 grade steel. Check for flexure and shear capacity only. | 16 | L4 | 2 | 2,3 |

Unit – III

- | | | | | |
|--|----|----|---|---|
| 5. a) Explain the behavior and failure modes of beam-columns with reference to slenderness ratio and axial force. Illustrate with graphs. | 10 | L3 | 3 | 2 |
| b) Discuss (i) first order and second order moments in beam-columns, (ii) columns bent in single curvature and double curvature, with relevant sketches. | 10 | L3 | 3 | 2 |

P.T.O

6. Check the adequacy of ISHB 300 @ 618 N/m, when it is used as a column of an industrial building over a length of 8.0 m, treating the ends as hinged. The column has to carry a factored axial load of 465 kN and maximum positive moment of 110 kN-m and maximum negative moment of 40 kN-m. 20 L4 3 2,3

Unit – IV

7. a) Discuss the concept of force distribution and failure pattern in beams with web openings, with sketches. 10 L3 4 2
 b) Highlight the steps involved in the analysis of beams with perforated thin webs, with sketches. 10 L3 4 2
8. Design the top chord member of a Virendel girder for the following data: Factored moment = 40 kN-m, Factored shear = 35 kN, Factored axial force = 140 kN, $f_y = 240$ MPa, $f_u = 410$ MPa. 20 L4 4 2,3

Unit – V

9. a) Discuss the advantages of cold form steel structures. 06 L2 5 1
 b) Compute the compressive load carrying capacity of the section shown in Fig. 9(b). Take $f_y = 240$ MPa.

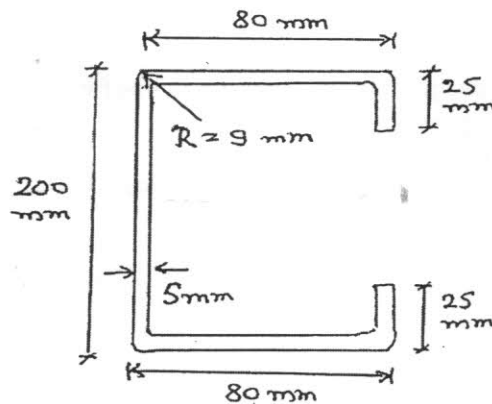


Fig. 9(b)

14 L3 5 2

10. a) Discuss the design principles of rounded tubular structures. 06 L2 5 1
 b) Evaluate the effective section modulus of the section shown in Fig. 10(b) and hence the maximum UDL it can carry when used as a simply supported beam over an effective span of 3.0 m. Take $f_y = 235$ MPa.

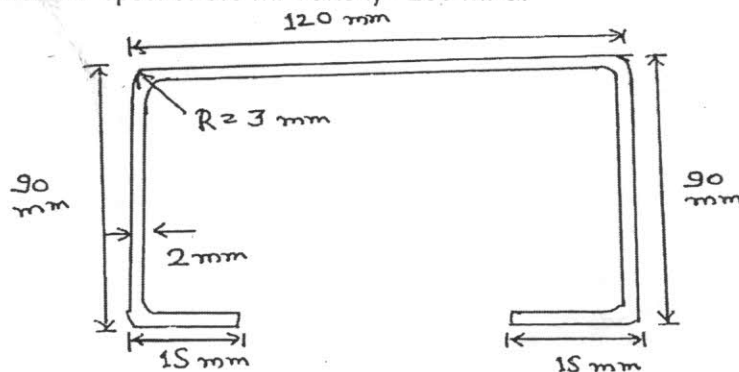


Fig. 10(b)

14 L3 5 2